SLUB Notes 2

2026.4.15

Slub Allocator主要结构体

struct kmem_cache

```
struct kmem_cache {
    struct kmem_cache_cpu __percpu *cpu_slab;
    slab_flags_t flags;
    unsigned long min_partial;
    unsigned int size;
    unsigned int object_size; // objsize
    struct reciprocal_value reciprocal_size;
    unsigned int offset;
#ifdef CONFIG_SLUB_CPU_PARTIAL
    unsigned int cpu_partial;
    unsigned int cpu_partial_slabs;
#endif
    struct kmem_cache_order_objects oo; // objperslab & pagesperslab

    /* Allocation and freeing of slabs */
    struct kmem_cache_order_objects min;
    gfp_t allocflags;
    int refcount;       /* Refcount for slab cache destroy */
    void (*ctor)(void *);
    ...
    const char *name;   /* Name (only for display!) */
    struct list_head list;  /* List of slab caches */
    ...
    struct kmem_cache_node *node[MAX_NUMNODES];
};
```

struct kmem_cache_cpu

```
struct kmem_cache_cpu {
    void **freelist;    /* Pointer to next available object */
    unsigned long tid;  /* Globally unique transaction id */
    struct slab *slab;  /* The slab from which we are allocating */
#ifdef CONFIG_SLUB_CPU_PARTIAL
    struct slab *partial;   /* Partially allocated frozen slabs */
#endif
    local_lock_t lock;  /* Protects the fields above */
#ifdef CONFIG_SLUB_STATS
    unsigned stat[NR_SLUB_STAT_ITEMS];
#endif
};
```

struct kmem_cache_node

```
struct kmem_cache_node {
    spinlock_t list_lock;

    unsigned long nr_partial;
    struct list_head partial;
#ifdef CONFIG_SLUB_DEBUG
    atomic_long_t nr_slabs;
    atomic_long_t total_objects;
    struct list_head full;
#endif
};
```

struct slab

```
struct slab {
    unsigned long __page_flags;

    union {
        struct list_head slab_list;
        struct rcu_head rcu_head;
#ifdef CONFIG_SLUB_CPU_PARTIAL
        struct {
            struct slab *next;
            int slabs;  /* Nr of slabs left */
        };
#endif
    };
    struct kmem_cache *slab_cache;
    /* Double-word boundary */
    void *freelist;     /* first free object */
    union {
        unsigned long counters;
        struct {
            unsigned inuse:16;
            unsigned objects:15;
            unsigned frozen:1;
        };
    };
    unsigned int __unused;

    atomic_t __page_refcount;
#ifdef CONFIG_MEMCG
    unsigned long memcg_data;
#endif
};
```